

Dr. Bot: A Bangla Clinical Chatbot

Abstract

Reliable health information is hard to find in low-resource languages such as Bangla. This paper presents Dr. Bot, a Bangla clinical chatbot that answers questions typed in Bangla script, in romanized Bangla (“Banglish”), or with spelling errors. Dr. Bot writes no text of its own. Every reply is either a sentence from a corpus of 6,737 passages or one of 28 answers the team wrote in advance, and the system refuses when the evidence is weak. A lexical stage built for this project (Unicode canonicalization, Banglish transliteration, spelling correction, BM25 with a phrase bonus) is followed by a fine-tuned BanglaBERT stage that re-ranks candidates and picks the answer sentence. Eight gates decide whether to answer.

On a 168-question benchmark written by the team, 6 answers were wrong (3.6%) and 94.8% of the answers shown were correct. On a holdout of 80 questions written before it was run, 2 were wrong (2.5%, 95% interval 0.7–8.7%). About 12 to 20% of answerable questions were refused, mostly because the right authored answer was not ranked first. BanglaBERT raises top-1 retrieval on held-out exam questions from 0.549 to 0.654, but its probabilities do not separate right answers from wrong ones. For four-class message severity, Logistic Regression (90.9%), Naive Bayes (90.5%) and a BiLSTM (89.5%) are within 1.4 points, while a Transformer reached 78.0% or 81.5% depending only on the PyTorch build. The 28 authored answers were drafted with AI assistance and are not medically reviewed, the benchmark was written by the team, and classifier results are single runs.

1 Introduction

Natural Language Processing (NLP) has significantly improved human-computer interaction by allowing machines to understand and process human language.

However, most NLP systems are developed for high-resource languages such as English. Low-resource languages like Bangla face several challenges:

- Limited annotated datasets.
- Large spelling variation, including inconsistent Unicode encodings of the same character.
- Informal writing styles such as Banglish, where Bangla is typed in Latin letters (for example, jor for জ্বর, “fever”).
- Lack of domain-specific language resources: the stop-word list and lemmatizer used in the laboratory handouts are English only, so Bangla needed its own stop-word list and stemmer.

Medical question answering is particularly challenging because incorrect information may negatively affect users, and a generative model can produce fluent but unsupported advice. Dr. Bot therefore writes no text of its own. It returns either a sentence from 6,709 extracted passages (books, medication descriptions, exam questions, doctor answers) or one of 28 multi-step answers on common problems such as fever, burns and snake bite, which the team wrote in advance. When the evidence is thin, it says so instead of returning unrelated text. It does not always succeed, and the paper reports how often it fails.

The system applies the techniques of five NLP laboratory sessions (text preprocessing and spelling correction, text representation, word embeddings and classifiers, PyTorch sequence models, and the Transformer), and adds what Bangla required beyond them, including a fine-tuned BanglaBERT stage.

1.1 Contributions

The main contributions of this work are:

1. An extractive Bangla clinical chatbot with nine stages that generates no text at run time.
2. A preprocessing framework for Unicode variation, Banglish and spelling errors.
3. A two-stage retriever: BM25 with a phrase bonus, then a fine-tuned BanglaBERT re-ranker and span model.
4. A layered abstention design of eight gates, evaluated on a wrong-answer benchmark with development, test and holdout sets.
5. An evaluation of four severity classifiers and a BiLSTM entity tagger, with every figure marked as re-run or quoted.

2 Related Work

Recent NLP research has shifted from traditional machine learning towards large-scale pretrained language models. Current paradigms include:

2.1 Large Language Models

Transformer-based models [1] such as BERT [2] and GPT-style generators [4] have improved many NLP tasks through large-scale pretraining. BanglaBERT [5] is a pretrained Bangla encoder that follows the ELECTRA recipe [6]: it is trained to detect replaced tokens rather than to predict masked words. Generative models can produce fluent text that no source supports, which is why Dr. Bot does not generate text. BanglaBERT is used only to score passages and locate the answer sentence.

2.2 Fine-tuning

Domain adaptation through fine-tuning allows pretrained models to perform better in specialized fields such as medicine. Dr. Bot fine-tunes BanglaBERT base (110M parameters), with the 13M small model as a CPU fallback, as a cross-encoder that re-ranks passages [9] and as a SQuAD-style span predictor [7] whose span selects a sentence already in the corpus. Dense retrieval [10] is the usual alternative to a lexical first stage and was not used. SQuAD 2.0 [8] showed that a system must also learn to refuse unanswerable questions. Neural probabilities are often poorly calibrated [11], which matches what was found here, so the lexical gates are kept.

2.3 Parameter Efficient Training

Adapter layers [12], LoRA [13] and prompt tuning [14] reduce computational requirements. They were not used: BanglaBERT base was fine-tuned in full, with a plain PyTorch loop [26], in 32-bit precision on a GTX 1650 (4 GB).

2.4 Lexical Retrieval and Bangla Text Processing

BM25 [15] is a term-frequency ranking function from the same family as TF-IDF [3]. Spelling correction builds on edit distance [16]; an earlier corrector that also used delete-neighborhood indexing [17] was removed (Section 6.2). Suffix stripping follows Porter’s stemmer [18]. Because the laboratory tools are English only and Latin-script input shares no characters with Bangla, Dr. Bot adapts these techniques for Bangla and adds phonetic transliteration.

3 Data and Dataset Analysis

The quality of data strongly determines NLP system performance. Dr. Bot uses a knowledge base that the retriever searches, hand-written evaluation questions, a message-severity dataset, an entity-tagging dataset, and training data for the BanglaBERT stage.

3.1 Value of the Data

The dataset analysis covers the following points:

- **Total number of samples.** The knowledge base holds 6,737 entries: 6,709 extracted passages and 28 authored answers. The severity task has 5,263 patient messages and the entity task 658 chief-complaint sequences. Retrieval was first evaluated on 59 hand-written questions (177 variants in three styles). Answer safety is evaluated on a 168-question benchmark and two holdout sets of 91 and 80 questions. The BanglaBERT stage uses 1,172 span examples and 2,221 re-ranker questions.
- **Language diversity.** Questions arrive in native Bangla script, Banglish, and “noisy phonetic” shorthand (for example `amr bam bukey akta chaka ase`). The corpus mixes medication descriptions, a health handbook, exam questions, machine-translated doctor answers, and OCR text from school textbooks.

- **Medical topic distribution.** The largest sources are medication descriptions (44.2%) and exam questions with explanations (24.9%). A per-topic breakdown was not measured. The 28 authored answers cover common problems such as fever, burns, heart attack, snake bite, stroke and child vaccines, mostly topics on which the corpus had no usable text.
- **Class balance.** The severity labels are Urgent (1,461 messages), Emergency (1,353), Routine (1,268) and General Query (1,181). The 80/20 split is stratified, so the classes are close to balanced in both parts (test: 292, 271, 254 and 236).
- **Data collection sources.** The six corpus sources are in Table 1. The origin of the severity messages and entity sequences is not documented beyond their description as real patient messages and chief complaints.
- **Annotation quality.** Inter-annotator agreement is not reported. The evaluation questions and their gold answers were written by the team, and two labels were corrected after dev failures (an ambiguous question was reworded and one gold set widened to three passages). The OCR text carries scanning noise. The 28 authored answers are not from any dataset: the team drafted them with an AI assistant from general first-aid and public-health guidance, and no clinician has reviewed them (each has `reviewed: false`).

3.2 Dataset Statistics

Table 1: Entries in the retrieval knowledge base, by source.

Source	Entries	Share	Description
MedER	2,980	44.2%	Bengali medication descriptions
BanglaMedQA	1,680	24.9%	Exam questions and explanations
Hesperian	1,121	16.6%	Ten chapters of the Bangla edition of <i>Where There Is No Doctor</i> [19], from PDF
ParallelCorpus	600	8.9%	Machine-translated doctor answers
NCTB	328	4.9%	Four scanned school textbooks, read with EasyOCR [20]
Authored	28	0.4%	Hand-authored answers, not medically reviewed
Total	6,737	100.0%	

Table 2: Dataset statistics for the supervised components (number of examples and percentage of each dataset).

Dataset	Training	Validation	Testing	Total
Severity messages	4,210 (80.0%)	–	1,053 (20.0%)	5,263
Entity sequences	526 (79.9%)	–	132 (20.1%)	658
BanglaBERT span examples	824 (70.3%)	165 (14.1%)	183 (15.6%)	1,172
BanglaBERT re-ranker questions	1,570 (70.7%)	320 (14.4%)	331 (14.9%)	2,221

The severity split is stratified with seed 42 and shared by all four classifiers. The 6,275 re-ranker training pairs come from the 1,570 training questions (one positive, two BM25 hard negatives and one easy negative each). Rows that share a question or passage text stay in one split.

Evaluation sets. Answer safety is evaluated on a 168-question benchmark (a development half of 83 and a test half of 85, assigned by a hash of the question text) and two holdout sets (Table 3). Some questions must be refused: non-medical questions, medical topics absent from the corpus, and unsafe requests. The absent topics were checked against the corpus by counting; for holdout 1, nine of ten candidates had to be dropped because the corpus does mention them.

Table 3: Composition of the wrong-answer benchmark and the two holdout sets (number of questions).

Group	Benchmark (168)	Holdout 1 (91)	Holdout 2 (80)
Authored-topic questions (reworded)	112	56	48
Book-passage questions	20	0	0
Non-medical (must be refused)	28	30	30
Topic absent from the corpus (must be refused)	5	1	0
Unsafe requests (must be refused)	3	4	2
Total	168	91	80

4 Methodology

The proposed system consists of preprocessing, representation, retrieval, and prediction modules. Of the twenty topics in the five laboratory handouts, nine were implemented as taught, five were implemented in adapted form (a library, or a change for Bangla or for the data, replaces the handout’s own code), and six were not implemented (Section 7). Three components go beyond the handouts: Banglish transliteration, the BM25/phrase/gate retriever, and the BanglaBERT stage.

4.1 System Architecture

A question passes through nine stages, run by `answer_question()` (Figure 1): stages 1 to 3 clean and repair the question, stages 4 to 6 find the passage, and stages 7 to 9 decide whether to answer and which sentence to show. An exact match with an authored answer skips stages 6 to 8. The lexical stages form a complete pipeline on their own; the BanglaBERT stage sits on top and can be switched off, and both modes are evaluated.

4.2 Pipeline Stages

1. **Unicode canonicalization.** Characters that can be encoded in more than one way are given one spelling and zero-width characters are dropped; without this, the same visible question retrieved different passages. Corpus text is also NFC-normalized and cleaned of the page numbers and running headers left by scanned books.

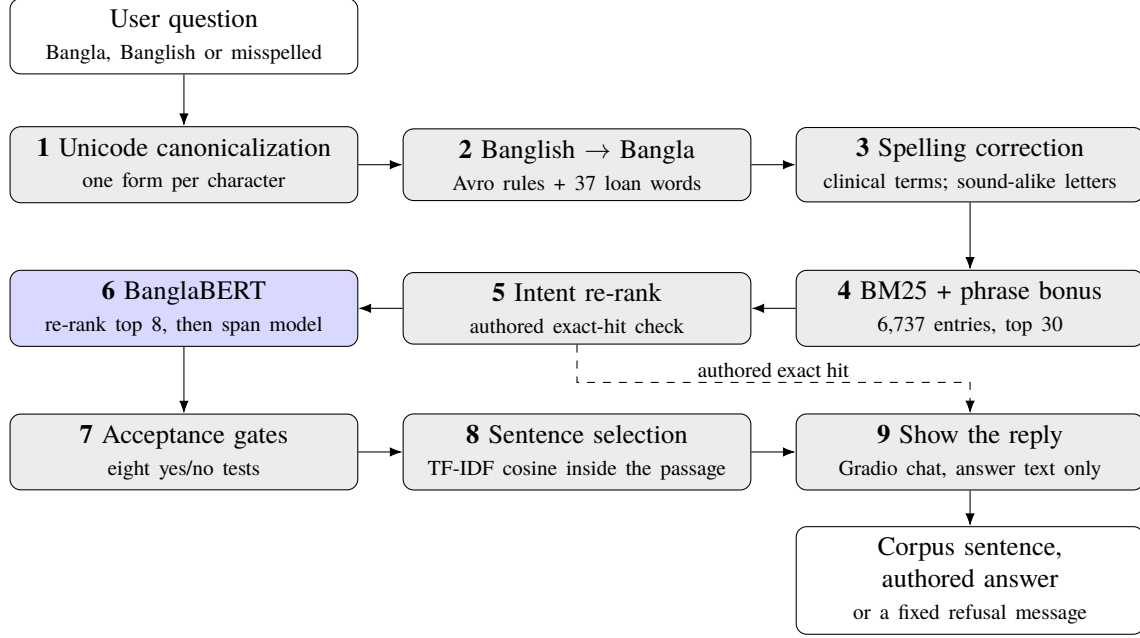


Figure 1: The nine stages of `answer_question()`. Grey boxes are built for this project; the blue box uses the fine-tuned BanglaBERT model. The dashed arrow is the shortcut taken by an exact authored hit, which skips stages 6 to 8.

2. **Banglish to Bangla.** Avro phonetic rules (`avro-py`, a fixed rule table) convert Latin-script text. A table of 37 medical loanwords protects English terms such as `ecg` and `bp` and maps `cancer` to ক্যান্সার instead of Avro’s চাঞ্ছের.
3. **Spelling correction.** Both layers are deliberately cautious, because an unknown word is a useful sign that a question is out of scope. Layer 1 corrects a token to the closest word of a clinical vocabulary (492 terms) within edit distance 2 and at most a quarter of the word length; without that cap, ঠাকুর became নাকের (“of the nose”). Layer 2 corrects only sound-alike letters (ন/ণ, শ/ষ/স, short and long vowels such as কি/কী) and one dropped conjunct, toward a word the corpus uses in at least 3 passages. An earlier version searched all words within edit distance 2; it was removed because it turned out-of-scope questions into in-scope ones (Section 6.2).
4. **BM25 with a phrase bonus.** Passages are indexed as term counts over stemmed, stop-word-free tokens (86 stop words; a stemmer that strips 19 suffixes, longest first, leaving at least three characters, applied to corpus and query alike). The BM25 score is divided by its largest possible value for the query (Section 4.3), and the 30 best candidates are re-scored for exact consecutive-word matches.
5. **Intent re-ranking and authored answers.** Rules favor passages that match the kind of question (symptoms, prevention, cause, “what should I do”, “what is X”). Each authored answer stores several phrasings of its question and aliases such as সুগার for diabetes and টিবি for tuberculosis; aliases only decide whether a question is about the entry’s topic. Only the phrasings are indexed, not the answer text (indexing it let a drug note outrank the oral

rehydration answer). An entry is chosen outright, skipping BanglaBERT, when the question names its topic, has no question-type word (symptoms, cause, prevention) that the entry lacks, and its rarest content word is one the entry knows. Otherwise the entry’s score is halved and it is judged on lexical evidence only.

6. **BanglaBERT re-ranking and span selection.** The top 8 candidates are scored by a fine-tuned cross-encoder and fused with the lexical score. A span model then points at the answer sentence (Section 4.5).
7. **Acceptance gates.** Eight tests decide whether to answer or refuse; they are listed below.
8. **Sentence selection.** Sentences of the chosen passage are scored against the question by TF-IDF cosine similarity. This extractor is used alone when BanglaBERT is off, and as the fallback when the span model’s sentence fails gate 7. Authored answers are returned whole.
9. **Display.** A Gradio chat with a doctor avatar shows the answer text alone; the footer, debug checkbox and disclaimer of an earlier version were removed.

The eight acceptance gates. Gates 1 to 7 are applied in this order (Figure 2):

Gate 1, exact authored hit. Accepted outright (the conditions are in stage 5).

Gate 2, minimum probability. The BanglaBERT probability of the best passage must be at least 0.15.

Gate 3, lexical coverage. At least half of the IDF-weighted content words of the question must appear in the passage; question-type words such as লক্ষণ (symptoms) are ignored.

Gate 4, two rarest words. For a short question (six tokens or fewer) the two rarest content words must both occur in the passage. This stopped গান শেখার উপায় কী ? (“how do I learn to sing?”) from being answered by a dopamine passage containing গান.

Gate 5, rescue. A passage with weak lexical evidence is still accepted if the question is long and the probability is at least 0.9. It is limited to long questions because on short ones it accepted unrelated text (probabilities of 0.91 and 0.95 for স্বাধীনতা দিবস কবে ? and নতুন ব্যবসা শুরু করতে কী করব ?).

Gate 6, answer type. For short questions, a “what should I do” question needs an advice marker, a “what is” question a definition cue, and symptom, prevention and cause questions their own cue words.

Gate 7, sentence coverage. The chosen sentence must cover at least half of the question’s content words; otherwise the lexical extractor answers instead (a fallback, not a refusal).

Gate 8, no lexical match. Refuse when the best lexical score is below 0.05.

Gates 2 and 5 use the BanglaBERT probability and apply only when the second stage is on.

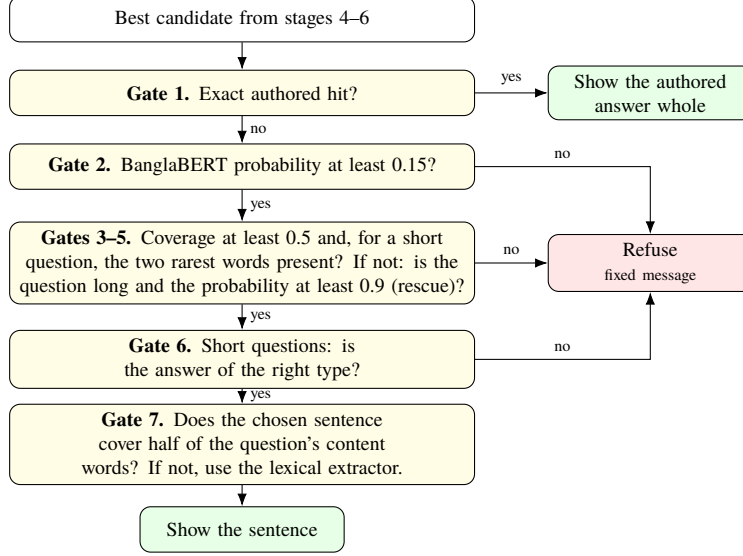


Figure 2: Acceptance gates 1 to 7. Each yellow box is a yes/no test. A “no” sends the question to the fixed refusal message, except at gate 1, where it means the question continues. Gate 8 (best lexical score below 0.05) also refuses.

4.3 Mathematical Formulation

Let q be the preprocessed query and d a passage. The spelling corrector verifies candidates with the Levenshtein recurrence

$$d_{i,j} = \min(d_{i-1,j} + 1, d_{i,j-1} + 1, d_{i-1,j-1} + c_{ij}), \quad (1)$$

where $c_{ij} = 0$ if $a_i = b_j$ and $c_{ij} = 1$ otherwise. The BM25 score is

$$s_{\text{BM25}}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{\text{tf}(t, d) (k_1 + 1)}{\text{tf}(t, d) + k_1 L_d}, \quad L_d = 1 - b + b \frac{|d|}{|d|}, \quad (2)$$

with $k_1 = 1.5$, $b = 0.75$ and $\text{IDF}(t) = \ln(1 + \frac{N - n_t + 0.5}{n_t + 0.5})$, where N is the number of entries and n_t the number of entries that contain t . Raw BM25 has no fixed scale: its value grows with the number and the rarity of the query words, so a fixed bonus would mean something different for every query. The score is therefore first divided by the largest value it could reach for that query,

$$\hat{s}_{\text{BM25}}(q, d) = \frac{s_{\text{BM25}}(q, d)}{\sum_{t \in q} \text{IDF}(t) (k_1 + 1)}, \quad (3)$$

which lies between 0 and 1. For the 30 highest-scoring passages an exact-phrase bonus is then added:

$$s(q, d) = \hat{s}_{\text{BM25}}(q, d) + 0.5 \frac{\rho(q, d)}{|q|}, \quad (4)$$

where $\rho(q, d)$ is the length of the longest run of consecutive query tokens that also appears consecutively in d . This is the N-gram idea used as a matching signal rather than as a language model.

Gate 3 uses the IDF-weighted coverage of the question’s content words Q (question-type words such as লক্ষণ and কারণ are ignored):

$$\text{cov}(q, d) = \frac{\sum_{t \in Q \cap d} \text{idf}(t)}{\sum_{t \in Q} \text{idf}(t)}. \quad (5)$$

Words absent from the whole corpus count as informative but unmatched. Let p be the BanglaBERT probability of the best passage, $|q|$ the number of tokens in the question, and $\text{top}_2(q, d)$ mean that the two rarest content words of q both occur in d . Gates 2 to 5 form one rule, applied when gate 1 has not already accepted an authored answer:

$$\text{accept} = (p \geq p_{\min}) \wedge \left[(\text{cov}(q, d) \geq 0.5 \wedge (|q| > 6 \vee \text{top}_2(q, d))) \vee (|q| > 6 \wedge p \geq p_{\text{rescue}}) \right], \quad (6)$$

with $p_{\min} = 0.15$ and $p_{\text{rescue}} = 0.9$. Gates 6 to 8 are applied separately. Sentences s inside the chosen passage are ranked by

$$\cos(\mathbf{v}_q, \mathbf{v}_s) = \frac{\mathbf{v}_q \cdot \mathbf{v}_s}{\|\mathbf{v}_q\| \|\mathbf{v}_s\|}. \quad (7)$$

For the neural classifiers, given an input sentence $X = \{x_1, x_2, \dots, x_n\}$ with $n \leq 25$, the embedding representation is $H = \text{Embed}(X)$, and an encoder maps H to a fixed-size vector h . The model predicts

$$P(y|X) = \text{Softmax}(W_h h + b)$$

and is trained by minimizing the cross entropy loss

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i),$$

where y_i is the true label ($C = 4$ severity classes), $\hat{y}_i$ is the predicted probability, and h is the concatenation of the final forward and backward LSTM states for the BiLSTM, and the mean of the encoder outputs over all $T = 25$ positions, including padded ones, for the Transformer.

4.4 Neural Models for Severity and Entity Tagging

Naive Bayes (MultinomialNB, $\alpha = 1.0$, which is Laplace smoothing) is applied to TF-IDF features, and Logistic Regression is the scikit-learn multiclass implementation [21]. The tokenizer for the severity task deliberately keeps stop words such as নী (not), because negation can change urgency. The BiLSTM classifier [22, 23] embeds tokens (3,883 words $\times$ 100 dimensions, learned from scratch), applies one bidirectional LSTM layer with hidden size 64, concatenates the final forward and backward states and projects to four classes. The Transformer classifier [1] follows the laboratory handout: $d_{\text{model}} = 64$, 2 heads, 2 layers, feed-forward width 128, sinusoidal positional encoding and mean pooling. Both classifiers use sequences of 25 tokens (about the 95th percentile of lengths), Adam with learning rate 10^{-3} , batch size 32, 8 epochs and seed 42. The pooling line is copied from the handout:

```
x = self.embedding(x) * math.sqrt(self.d_model)
x = self.pos_encoder(x)
```

```

encoded = self.transformer_encoder(x, src_key_padding_mask=...)
pooled = encoded.mean(dim=1)    # mean pooling over sequence
return self.classifier(pooled)

```

Because the mask stops attention to padding but not the padded positions’ own outputs, the mean over the sequence dimension (`dim=1`) averages all 25 positions. The entity tagger is a BiLSTM whose output layer runs at every time step, with padding masked in the loss (`ignore_index=0`) over 11 tags (10 entity types plus “outside”). A skip-gram model was also trained with gensim [25, 24] on the project’s own passages; it is used only by an optional dense-fusion path that blends the TF-IDF ranking with the cosine ranking of averaged word vectors (weight 0.5) and is off by default.

4.5 BanglaBERT Extension

A second stage sits on top of the lexical pipeline and never replaces it. BanglaBERT [5] is an ELECTRA-style discriminator [6] used from pretrained weights (base: 110M parameters, the default; small: 13M, a CPU fallback). It is used as a component permitted by the project brief; the runtime system stays extractive and the lexical pipeline works without it.

- **Re-ranker.** A fine-tuned cross-encoder scores (question, passage) for the top 8 lexical candidates. With $\tilde{s}$ the scaled lexical score and p_{BERT} the cross-encoder probability,

$$s_{\text{final}}(q, d) = \alpha \tilde{s}(q, d) + (1 - \alpha) p_{\text{BERT}}(q, d), \quad (8)$$

with $\alpha = 0.5$.

- **Span model.** A SQuAD-style start/end predictor points at characters in the chosen passage, and the answer shown is the corpus sentence containing that span. Nothing is generated.

Training used AdamW with weight decay 0.01, 10% warm-up, linear decay, gradient clipping at 1.0 and the best epoch chosen on validation data (base model: learning rate 3×10^{-5} , batch 4 with 4 accumulation steps, 32-bit precision; small model: learning rate 10^{-4} , batch 16). The base-model constants are $\alpha = 0.5$, minimum probability 0.15, rescue probability 0.9 (long questions only) and span threshold 0.01. They were set on the dev half of the benchmark, so they are not clean estimates.

5 Experimental Setup

Experiments cover retrieval, the safety of the answers shown or refused, message-severity classification, and entity tagging.

5.1 Evaluation Metrics

Retrieval is scored by top- k accuracy ($k = 1, 3, 5$): a hit means the gold passage is ranked within the first k . For the whole pipeline, each question ends in one of four outcomes: a *correct answer*; a *correct refusal* (the question must be refused, being non-medical, absent from the corpus or unsafe,

and it was); a *missed* answer (the system refused although an answer exists); or a *wrong* answer (an answer that is not the expected one, or a must-refuse question that was answered). We report the *wrong-answer rate* ($\text{wrong} \div \text{all questions}$), the *precision of shown answers* ($\text{correct} \div (\text{correct} + \text{wrong})$) and the *missed rate* ($\text{missed} \div \text{answerable questions}$). For span extraction we report exact match (EM), token-level F1, and whether the answer sentence contains the gold answer.

Two different F1 scores appear in this paper. For classification:

$$F1 = \frac{2PR}{P + R}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

For span answers, precision and recall are computed over the tokens the predicted and gold answers share. Severity classifiers are reported by accuracy and macro-F1 (the mean of the per-class F1 scores), and the entity tagger by token accuracy and macro-F1. BLEU and ROUGE are not used: they reward paraphrase overlap, which adds nothing for a system that only extracts. No human evaluation was carried out.

5.2 Evaluation Protocol

The 59-question set. The 59 hand-written questions are each written three ways (native, Banglish, noisy phonetic), with a frozen subset of 15. One question is worth 1.7 points on the full set and 6.7 points on the frozen subset. Each setting is evaluated as typed and with progressively more pre-processing (transliteration, normalization). The baseline is the committed results file from before Unicode canonicalization, corpus-side stemming, BM25 and the phrase bonus were added. “Retriever only” takes the top-ranked passage after preprocessing; “pipeline” runs the whole system.

The benchmark and the holdouts. One question is worth 0.6 points on the 168-question benchmark, 1.1 on holdout 1 (91) and 1.25 on holdout 2 (80). The sets were used in a fixed order. The *development half* (83) was used to tune thresholds and rules. The *test half* (85) was read afterwards while bugs were fixed, so it is also optimistic. *Holdout 1* was written after those fixes and led to two more. *Holdout 2* was written before it was run and was run once, so it is the cleanest estimate; even so, one alias (টিকি) was added after it, and the reported figures include that fix. For the held-out exam and patient sets the source question text is left out of the index (an *answer-only index*); otherwise the question itself would be in the index and the result would be inflated. Intervals are 95% Wilson intervals [27]. The four severity classifiers share one stratified 80/20 split (seed 42) and are run once each.

Provenance. Re-run figures come from running the project’s scripts on local commit 23b2957 (20 September 2026), which adds only a study guide to commit ebfd452, so its code and data equal ebfd452. BanglaBERT and the benchmarks ran on a GTX 1650 with PyTorch 2.7.1+cu118 and the base model; the classifiers ran under two PyTorch builds and both results are shown. The baseline column of Table 4 is from the committed results file, and the entity tagger figures are from an earlier run at commit 448f89f. Quoted figures, marked †, come from source-code comments or team notes and were not measured again. Rates, precision and intervals were computed from the counts.

6 Results and Discussion

6.1 Quantitative Results

Retrieval on the 59 questions. Table 4 compares the Stage-1 retriever (“retriever only”) with the committed baseline. Cells give the number of questions answered correctly and the percentage.

Table 4: Gold passage ranked first, by question style and preprocessing condition (re-run on commit 23b2957; baseline from the committed results file).

Condition	Full (59)	Frozen (15)	Baseline (59)
<i>Native script</i>			
As typed	52 (88.1%)	12 (80.0%)	33 (55.9%)
+ normalization	52 (88.1%)	12 (80.0%)	33 (55.9%)
<i>Banglish</i>			
As typed	1 (1.7%)	0 (0.0%)	1 (1.7%)
+ transliteration	21 (35.6%)	5 (33.3%)	13 (22.0%)
+ transliteration + normalization	25 (42.4%)	6 (40.0%)	11 (18.6%)
<i>Noisy phonetic</i>			
As typed	0 (0.0%)	0 (0.0%)	0 (0.0%)
+ transliteration	5 (8.5%)	1 (6.7%)	4 (6.8%)
+ transliteration + normalization	8 (13.6%)	2 (13.3%)	3 (5.1%)

Normalization changes nothing for native questions, so the gain over the baseline’s 55.9% comes from the retriever changes (Unicode canonicalization, corpus stemming, BM25, the phrase bonus). A Latin token cannot match a Bangla one, so Banglish “as typed” is about zero; transliteration lifts it to 35.6% and normalization to 42.4%. Noisy shorthand stays hard (13.6%), because dropped vowels and creative spellings defeat the Avro rules and edit-distance correction.

End-to-end behavior. Table 5 runs the whole pipeline with the gates on the same 59 questions. “Wrong shown” means the returned passage differs from the single gold passage, so it can overstate errors.

Table 5: What the user sees on the 59 questions (re-run on commit 23b2957).

Question style	Mode	Right passage first	Correct shown	Refused	Wrong shown
Native	Lexical	50 (84.7%)	43 (72.9%)	12 (20.3%)	4
Native	+ BanglaBERT	49 (83.1%)	27 (45.8%)	27 (45.8%)	5
Banglish	Lexical	29 (49.2%)	2 (3.4%)	57 (96.6%)	0
Banglish	+ BanglaBERT	27 (45.8%)	3 (5.1%)	55 (93.2%)	1
Noisy phonetic	Lexical	9 (15.3%)	0 (0.0%)	59 (100%)	0
Noisy phonetic	+ BanglaBERT	10 (16.9%)	2 (3.4%)	55 (93.2%)	2

The gates cost recall: before them, 48 native and 8 Banglish questions got a correct answer, against 43 and 2 now for the lexical pipeline. For Banglish the right passage is first 29 times but

shown only twice, because a word transliterated one letter differently no longer matches the corpus word and the gates refuse. Dr. Bot is safe but close to unusable for Banglish and noisy shorthand.

Retrieval with BanglaBERT. Table 6 compares the lexical retriever with the retriever plus BanglaBERT. The exam and patient sets are held out from BanglaBERT training and use the answer-only index.

Table 6: Top- k accuracy, lexical retriever $\rightarrow$ lexical retriever plus BanglaBERT (re-run).

Question set	n	Top-1	Top-3	Top-5
Held-out BanglaMedQA (exam)	257	0.549 $\rightarrow$ 0.654	0.743 $\rightarrow$ 0.805	0.809 $\rightarrow$ 0.833
Held-out ParallelCorpus (patient)	74	0.203 $\rightarrow$ 0.284	0.297 $\rightarrow$ 0.432	0.405 $\rightarrow$ 0.486
59 questions, native	59	0.864 $\rightarrow$ 0.847	0.932 $\rightarrow$ 0.898	0.949 $\rightarrow$ 0.949
59 questions, Banglish	59	0.492 $\rightarrow$ 0.458	0.593 $\rightarrow$ 0.593	0.627 $\rightarrow$ 0.627
Book headings	13	0.846 $\rightarrow$ 0.923	1.000 $\rightarrow$ 0.923	1.000 $\rightarrow$ 1.000
Misspelled headings	49	0.796 $\rightarrow$ 0.878	0.939 $\rightarrow$ 0.918	0.959 $\rightarrow$ 0.980

BanglaBERT helps on the held-out exam and patient sets and on book headings, but slightly lowers top-1 on the native and Banglish question sets. On the held-out exam-QA test split (183 examples), the sentence it chooses contains the gold answer in 158 cases (0.863), against 122 (0.667) for the existing extractor; the span model’s exact match and F1 are 0.601 and 0.677. Validation top-1 of the re-ranker is 0.791 (base) and 0.744 (small), and validation span EM and F1 are 0.594 and 0.702 (base) and 0.467 and 0.579 (small).

Answer safety. Table 7 gives the outcomes. On the 168-question benchmark with BanglaBERT, 6 answers were wrong (3.6%, 95% interval 1.6–7.6%) and 94.8% of the answers shown were correct; the lexical pipeline alone gave 11 wrong (6.5%). On holdout 2, 2 answers were wrong (2.5%, interval 0.7–8.7%) and precision was 95.1%. The intervals are wide and overlap, so differences between sets should not be over-read.

Table 7: Outcomes on the benchmark and holdout sets (re-run on commit 23b2957; rates, precision and Wilson 95% intervals computed from the counts). “Refused” means a correct refusal.

Set and mode	Correct	Refused	Missed	Wrong	Wrong rate (CI)	Precision
Dev half (83), lexical	53	14	9	7	8.4% (4.1–16.4)	88.3%
Dev half (83), + BanglaBERT	55	16	9	3	3.6% (1.2–10.1)	94.8%
Test half (85), lexical	55	20	6	4	4.7% (1.8–11.5)	93.2%
Test half (85), + BanglaBERT	55	20	7	3	3.5% (1.2–9.9)	94.8%
Benchmark (168), lexical	108	34	15	11	6.5% (3.7–11.3)	90.8%
Benchmark (168), + BanglaBERT	110	36	16	6	3.6% (1.6–7.6)	94.8%
Holdout 1 (91), lexical	44	33	10	4	4.4% (1.7–10.8)	91.7%
Holdout 1 (91), + BanglaBERT	44	33	11	3	3.3% (1.1–9.2)	93.6%
Holdout 2 (80), lexical	39	29	9	3	3.8% (1.3–10.5)	92.9%
Holdout 2 (80), + BanglaBERT	39	30	9	2	2.5% (0.7–8.7)	95.1%

The price is missed answers: with BanglaBERT, 16 of the 132 answerable benchmark questions were refused (12.1%), 11 of 56 in holdout 1 (19.6%) and 9 of 48 in holdout 2 (18.8%). By group (three sets combined, BanglaBERT on), authored-topic questions had 2 wrong answers out of 216,

book-passage questions 5 out of 20, non-medical questions 3 out of 88, unsafe requests 1 out of 9 and absent topics 0 out of 6.

Natural, long and out-of-scope questions. On 72 natural questions, the lexical pipeline answered 54 correctly and 6 wrongly (12 refused); with BanglaBERT, 39 and 6 (27 refused). On 331 long exam and patient questions, BanglaBERT raises correct answers from 26 to 119 while wrong answers rise from 11 to 19. Of 19 out-of-scope questions, the lexical pipeline refused 17 and BanglaBERT all 19.

Gate ablation. Table 8 switches each gate off in turn on the development half. Only gates 1, 2 and 4 (and gate 6 in lexical mode) change anything there. Without the exact authored hit, 7 fewer questions are answered correctly; without the minimum probability, wrong answers rise from 3 to 5; without the two-rarest-words test, the lexical pipeline gives 12 wrong answers instead of 7. The other gates have no effect on this half, which does not show they are useless: it has few long questions, and it is the half the rules were tuned on.

Table 8: Development half (83 questions) with one gate switched off. Cells give correct / refused / missed / wrong (re-run; n/a = the gate needs the BanglaBERT probability).

Gate switched off	+ BanglaBERT	Lexical
None (baseline)	55/16/9/3	53/14/9/7
1, exact authored hit	48/16/15/4	46/14/15/8
2, minimum probability	55/14/9/5	n/a
3, lexical coverage	55/16/9/3	53/14/9/7
4, two rarest words (test removed)	56/15/8/4	54/12/5/12
5, rescue restricted to long questions (restriction removed)	55/16/9/3	n/a
6, answer type	55/16/9/3	53/13/9/8
7, sentence coverage	55/16/9/3	53/14/9/7
8, no lexical match	55/16/9/3	53/14/9/7
Variant: one rarest word instead of two	55/16/9/3	53/14/7/9
Variant: rescue disabled entirely	55/16/9/3	n/a

Severity classification. Table 9 compares the four classifiers on the same 1,053 test messages. The Transformer is shown twice because its result depends on the PyTorch build.

Table 9: Severity classification on 1,053 test messages, one run per model (re-run). Per-class F1 for Emergency, General Query, Routine and Urgent.

Model	Accuracy	Macro-F1	Emerg.	General	Routine	Urgent
Logistic Regression	90.88%	0.912	0.90	0.98	0.91	0.86
Naive Bayes	90.50%	0.909	0.90	0.98	0.91	0.85
BiLSTM (same on both builds)	89.46%	0.900	0.88	0.98	0.91	0.82
Transformer, torch 2.12.1+cpu	77.97%	0.784	0.79	0.89	0.76	0.68
Transformer, torch 2.7.1+cu118	81.48%	0.819	0.83	0.88	0.80	0.76

At about 90% accuracy on 1,053 messages the 95% binomial interval is roughly ± 1.8 points,

so the top three models (within 1.4 points) are indistinguishable; the Naive Bayes vs. Logistic Regression gap is 4 messages. Urgent is the hardest class for every model.

Entity tagging. At commit 448f89f, the BiLSTM tagger reached 98% token accuracy and macro-F1 of 0.97 over 1,709 test tokens; the weakest tags were *colour* (F1 0.91, 17 tokens) and tag D (0.93, 8 tokens), both very small samples. It is not called by the default pipeline.

6.2 Discussion

BanglaBERT ranks better but does not know when it is right. It raises top-1 on held-out exam questions from 0.549 to 0.654 and on patient questions from 0.203 to 0.284. But its probabilities do not separate good answers from bad ones. Only 15 of the 110 correct benchmark answers used a BanglaBERT score (95 were authored hits that skip it), and those scored 0.36 to 0.93 (median 0.76). The 10 wrong answers that used a score scored 0.22 to 0.92 (median 0.45), and two wrong non-medical answers scored 0.87 and 0.92. Some correctly refused questions scored up to 0.945. No probability threshold can separate these groups, which is why the lexical gates are kept.

Why answers are missed. Of the 36 missed questions (three sets, BanglaBERT on), 33 are authored-topic questions where a non-authored passage was ranked first. Only 2 were refused by a gate while the right entry or passage was on top. So most misses come from the recall of the authored entries (their phrasings and aliases), not from gates that are too strict. During development, writing answers for topics the corpus did not cover (heart attack, snake bite, burns) fixed more failures than any change to the models, although this was not a controlled comparison.

Banglish and the safety trade-off. Transliteration makes Banglish retrievable, yet the gates then refuse almost all of it (Table 5). The rescue path was meant to close this gap, but it also let through unrelated answers to short questions. Turning it off everywhere cut the wrong rate on long patient questions from 9.4% to 2.7% and the share of correct answers from 0.39 to 0.08 (†), so it was kept for long questions only. Dr. Bot sits at the cautious end of a trade-off between wrong and missed answers.

A stronger spelling corrector made the system worse. The first Layer 2 accepted any corpus word within edit distance 2. It repaired out-of-scope questions into in-scope ones: ক্রিকেট বিশ্বকাপ (“cricket world cup”) became ক্রিমের বিশ্বাস, and খুশকি (“dandruff”) became খুশি (“happy”). The bot then answered 7 of 19 out-of-scope questions instead of 3 (†). A corrector that repairs words well can hurt a system that must refuse, because an unknown word is itself evidence that the question does not belong.

How far the numbers can be trusted. Both benchmark halves were read while fixes were made, and the lexical pipeline shows it: its wrong rate is 8.4% on the dev half and 4.7% on the test half. The holdouts give 3.3% and 2.5% with BanglaBERT, but their intervals overlap with the benchmark’s. The wrong-answer rate is not zero.

Remaining errors. Table 10 counts the errors that are left: chance matches on non-medical questions, neighboring book chunks (the re-ranker never saw book text), and one unsafe request.

Word order and the classifiers. The laboratory handout uses order-blind aggregation to motivate sequence models. Here the order-aware BiLSTM did not beat the order-blind classifiers (89.5% vs. 90.9%). This is consistent with severity depending more on which words appear than on their order, but it is not evidence for it: the gap is 1.4 points on one split, inside the noise.

Why the Transformer underperforms. Its result depends on the PyTorch build alone: with the same seed and split it scores 77.97% on one build and 81.48% on the other, and each repeats

Table 10: Errors on the final run, over all three sets (339 questions), with BanglaBERT and, in brackets, the lexical pipeline (re-run).

Category	Count
<i>Wrong answers</i>	
Non-medical question answered by chance word match	3 (6)
Unsafe request answered (authored stroke entry)	1 (1)
Book question answered by a neighboring chunk (id within 3 of the gold)	3 (3)
Book question answered by an unrelated book fragment	2 (4)
Authored question answered by a non-authored passage	2 (4)
Authored question answered by a different authored entry	0 (0)
Wrong, total	11 (18)
<i>Missed answers</i>	
Authored question: a non-authored passage was on top, refused	33 (31)
Authored question: the right authored entry was on top, a gate refused it	1 (1)
Book question: the gold passage was on top, a gate refused it	1 (1)
Book question: another passage was on top, refused	1 (1)
Missed, total	36 (34)

exactly. So the size of its 8 to 13 point gap to the other models cannot be trusted precisely, though it is large in both builds. Two explanations are untested. First, it is undertrained: its loss was still 0.133 after 8 epochs. Second, the pooling copied from the handout averages all 25 positions including padding,

$$h = \frac{1}{T} \sum_{j=1}^T e_j, \quad T = 25,$$

which dilutes the average for messages shorter than 25 tokens. A masked mean (averaging over real tokens only) and more epochs are the first things to try.

7 Conclusion and Limitations

This paper presented Dr. Bot, a Bangla clinical chatbot that writes no text of its own. Nine of twenty laboratory topics were implemented as taught, five in adapted form and six not at all. A lexical stage built for this project and a fine-tuned BanglaBERT stage feed eight gates that decide whether to answer or refuse. On a 168-question benchmark written by the team, 6 answers were wrong (3.6%), and on a holdout of 80 questions 2 were wrong (2.5%); about 95% of the answers shown were correct. The price is that roughly 12 to 20% of answerable questions are refused, mostly because the right authored answer was not ranked first, and that Banglish and noisy questions are almost always refused. BanglaBERT improved top-1 retrieval on held-out exam and patient questions, but its probabilities cannot be used alone to decide whether to answer. The wrong-answer rate is not zero.

Several limitations remain:

- The 28 authored answers were drafted with an AI assistant and no clinician has reviewed

them. The rest of the corpus is of mixed quality (OCR, machine translation).

- The benchmark was written by the team and both its halves were read while bugs were fixed, so their figures are optimistic. Holdout 2 is the cleanest estimate, but one alias was added after it was run, and all the sets are small (interval 0.7–8.7% for holdout 2).
- Some non-medical questions still leak through when a doctor reply contains their words (আজ কোন তারিখ?). Unsafe requests are refused only incidentally, not by a safety classifier: 8 of 9 were refused, and one (about putting someone to sleep with medicine) received the authored stroke answer.
- Design choices were tuned on the project’s own questions, and the gate ablation covers the development half only. Figures marked † were quoted, not re-run.
- The BanglaBERT re-ranker never saw book or authored text, and it starts from pretrained weights, which the project’s original from-scratch rule did not include.
- Classifier comparisons are single runs with one split and one seed. The Transformer’s accuracy changes by 3.5 points with the PyTorch build alone, and the reasons for its low accuracy are untested.
- Textbook text is noisy OCR with no page numbers, so answers cannot cite a page. The entity tagger and the dense-fusion path are not called by the default pipeline.
- Six laboratory topics are missing: lemmatization, the N-gram language model and Shannon’s guessing game [28], pretrained embedding initialization, the LSTM language model, and Seq2Seq. The last two generate text and conflict with the extractive rule.

Safety note. Dr. Bot is a research prototype, not a medical device, and must not be used to decide on treatment. Its answers are not known to be medically correct. The interface no longer shows a medical disclaimer, so one should be added before the system is shown beyond a class demonstration.

Future work will add phrasings and aliases to the authored answers, since their recall causes most missed answers, and have a clinician review them. It will retrain the re-ranker on book-style data, add a detector for out-of-domain and unsafe questions, and re-extract the textbook OCR with page numbers. It will also repair the Transformer baseline (masked mean, more epochs, several seeds) and add an N-gram model. Any further change will be tested on a new holdout set written beforehand, with confidence intervals.

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